Physics - Graduate Level Assessment 6 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. What is the force between two positive charges of 100μ coulombs and 200μ coulombs, respectively, when they are separated a distance of 3 centimeters?

   (a) 5 x 10^5 nt

   (b) 1 x 10^5 nt

   (c) 7 x 10^5 nt

   (d) 3 x 10^5 nt

   (e) 6 x 10^5 nt

2. The direction of natural processes is from states of

   (a) lower order to higher order

   (b) chaos to equilibrium

   (c) higher order to lower order

   (d) disorder to organization

   (e) equilibrium to higher order

3. Any atom that emits an alpha particle or beta particle

   (a) changes its atomic number but remains the same element.

   (b) becomes an atom of a different element always.

   (c) remains the same element and isotope.

   (d) becomes a different element with the same atomic mass.

   (e) becomes an atom of a different element sometimes.

4. The separation of the headlights on a car is about 150 cm. When the car is 30 meters away, what is the separation of the images of the lights on the retina?

   (a) 4 mm

   (b) .833 mm

   (c) 1 mm

   (d) 1.5 mm

   (e) 0.5 mm

5. A hoist raises a load of 330 pounds a distance of 15 feet in 5 seconds. At what rate in horsepower is the hoist working?

   (a) 2.5 hp

(b) 2.7 hp

(c) 1.8 hp

(d) 1.2 hp

(e) 3.0 hp

True / False (10 marks)

Answer True or False

6. True or False: The work done to stretch the steel wire by 0.50 mm with a force of 100 N is 0.01 N · m.

7. True or False: The refractive index of the glass for a -10.00 diopter biconcave lens with 12 cm radii of curvature is 2.0.

8. True or False: A torque of -15 lb-ft is sufficient to stop a disk rotating at 1200 rev/min in 3.0 minutes.

9. True or False: The center of mass of a uniformly solid cone with base diameter $2a$ and height h is located at $rac34h$ from the base.

10. True or False: The speed of sound in air is significantly affected by the frequency of the sound wave being produced.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A curling stone with a mass of 1 slug travels along the ice and comes to rest after 100 ft due to a kinetic friction force of 1/2 lb. Calculate the initial velocity and the time taken to stop, providing detailed reasoning for your approach.

12. Discuss the relationship between the amplitude of the electric field and the Poynting vector in a light wave traveling through glass with an index of refraction of 1.5, given an electric field amplitude of 100 volts/meter. Calculate the magnitude of the Poynting vector and explain its physical significance in this context.

End of Paper